

Neuroscientific Frontiers — Module 05: Action —

Study Questions

1. How does Active Inference reconceptualize movement? What replaces the classical "motor command"? Why is this a radical departure from the standard motor neuroscience view?
2. How do spinal cord reflex arcs implement proprioceptive prediction error reduction? Why are classical reflexes (stretch reflex, withdrawal reflex) essential to this account?
3. What evidence supports the claim that motor cortex encodes intended trajectories rather than muscle activation patterns? How do decoding studies inform this debate?
4. What is the cerebellum's role as a forward model? How does it predict sensory consequences of motor actions before they occur?
5. How do climbing fibers signal motor prediction errors? What is the role of the inferior olive? How does this relate to error-driven learning more broadly?
6. How does long-term depression at parallel fiber-Purkinje cell synapses implement motor learning? Connect this to the Dirichlet parameter accumulation from the Advanced Theory track.
7. Why does practice improve motor skill? Explain in terms of forward model refinement. What specific neural changes occur with expertise?
8. How do the basal ganglia's direct, indirect, and hyperdirect pathways implement policy selection? Map each pathway to a specific Active Inference computation.
9. How does dopamine (D1/D2 balance) control the precision of motor policy selection? What happens when this balance is disrupted (Parkinson's vs. Huntington's)?
10. What is the supplementary motor area's role in action sequencing? How does it implement multi-step policy execution?
11. How does the readiness potential (Bereitschaftspotential) relate to policy precision? What does its pre-conscious timing mean for the concept of volition?
12. Does the RP undermine free will? How does Active Inference dissolve the Libet problem by reframing "decision" as precision accumulation?
13. Explain saccadic eye movements as Active Inference. What predictions are fulfilled? How does the superior colliculus generate target predictions?

14. How are saccades epistemic actions? What determines where the eyes move next in a natural scene? Connect this to expected free energy and epistemic value.
15. What is smooth pursuit? How does it require continuous prediction updating? What happens when the prediction lags behind the target?
16. How does sensory attenuation during self-generated movement support the active inference account? Why can't you tickle yourself?
17. Why do schizophrenia patients show reduced sensory attenuation? How does this relate to auditory hallucinations and passivity experiences? What does this reveal about the self-model?
18. How does Active Inference explain the force-matching illusion? What does this reveal about forward model calibration?
19. How does dystonia relate to inappropriate proprioceptive precision? How might precision-based interventions differ from traditional treatments?
20. Critically evaluate: Is the "predictions not commands" framework empirically distinguishable from classical motor control theories? What experiment could decisively resolve this debate?